(SSUET/QR/111)

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY

COMPUTER SCIENCE AND INFORMATION TECHNOLOGY DEPARTMENT

BS (INFORMATION TECHNOLOGY)

COURSE INFORMATION SHEET

Session:	Fall-2023
Course Title:	Software Engineering
Course Code:	CS-226
Credit Hours:	3+0
Semester	3rd
Pre-Requisites:	NONE
Instructor Name:	Mr. Umair Khan
Email and Contact Information:	umair.khan@ssuet.edu.pk, 34973414 Ext: 310
Office Hours:	Tuesdays (11:30 am – 1:00 pm) Wednesdays (02:30 pm – 4:00 pm)
Method of Teaching:	**Synchronous**/~~Asynchronous/ Hybrid/Blended~~

XXXXXXXX

COURSE OBJECTIVE:

The objective of this course is to:

- Establish a strong foundation in Software Engineering.
- Highlight the importance of Software Engineering through study of inclusive activities such as Requirement Engineering, System Modeling and Quality Management in detail.
- Create a basic capability of building a quality software through Software Engineering.
- Develop familiarity with tools and practices used in the industry.

COURSE OUTLINE:

Software and Software Engineering, Software Engineering Practices, Software Process Models, Agile Software Development, Scrum, Requirements Engineering Process, User Centered Design, System Modeling Techniques and UML, Software Architecture, Design Patterns, Software Reuse, Software Quality and Testing, Software Evolution, Configuration Management

COURSE LEARNING OUTCOMES (CLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	**Describe** key principles and activities of software engineering.	PLO_1 (Academic Education)	C2 (Understanding)
2	**Explain** the purpose of software engineering process elements.	PLO-2 (Knowledge for Solving Computing Problems)	C2 (Understanding)
3	**Apply** the software engineering knowledge to suggest an appropriate solution that meets the customer's requirements.	PLO-3 (Problem Analysis)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	4 (14%)	3 (8%)	3 (8%)
Assignments	4 (14%)	3 (8%)	3 (8%)
Mid	10 (36%)	10 (28%)	10 (28%)
Final	10 (36%)	20 (56%)	20 (56%)
Total	28 (28%)	36 (36%)	36 (36%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	**100%**	**100**

Recommended Books:

- Sommerville, I., 2016. *Software Engineering, 10/E Global*. Pearson Education. (IS-SE)
- Sommerville, I. *Engineering Software Products: Introduction to Modern Software Engineering*. Pearson Education. (IS-ESP)

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LECTURE PLAN

COURSE TITLE: SOFTWARE ENGINEERING **COURSE CODE: CS-226**

Week No.	Course Plan	Recommended Reading	Assessment (End of Week)
1	**Introduction To Software Engineering** • Professional Software Development • Software Engineering Ethics • Case Studies	IS-ESP Chapter 1	
2	**Software Processes** • Software Process Models • Process Activities • Plan-Driven and Agile Development • 12 Principle of Agile Manifesto	IS-SE Chapter 2	
3	**Agile Software Development** • Scrum • Scrum Practices and Roles • Kanban • Extreme Programming	IS-SE Chapter 3 IS-ESP Chapter 2 & 3	Quiz # 1
5	**Requirements Engineering I** • Functional And Non-Functional Requirements • Requirements Specification	IS-SE Chap 4	
6	**Requirements Engineering II** • Requirements Engineering Processes • Requirements Validation • Requirements Management	IS-SE Chap 4	Assignment #1
7	**System Modeling 1** • Context Models • Interaction Models • User-Centred Design	IS-SE Chapter 5	
8	**System Modeling II** • Structural Models • Behavioural Models • Model-Driven Engineering	IS-SE Chapter 5	Quiz # 2

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Week No.	Course Plan	Recommended Reading	Assessment Tools
10	**MIDTERM EXAMINATION**		
10	**Architectural Design** • Architectural Design Decisions • Architectural Views • Architectural Patterns • Application Architectures	IS-Chapter 6	
11	**Design And Implementation** • Object Oriented Design Using UML • Design Patterns **Introduction To Software Engineering Tools** • Microsoft Visio	IS-Chapter 7:	Assignment # 2
12	**Models of Software Engineering** • Software Reuse and application frameworks • Components and component model • Client-Server Computing • Software-as-a-Service	IS-SE Chapter 15,16 & 17 IS-ESP Chapter 5	

13	**Software Testing**
	• Development Testing
	  ○ Test-Driven Development
	  ○ Behaviour-Driven Development
	• Release Testing
	• User Testing
		IS-SE Chap 8	Quiz # 3

14	**Quality Management** • Software Quality • Code Reviews and Inspections • Quality Management and Agile Development • Code Smells	IS-SE Chap 24:	
15	**Software Evolution** • Evolution Process • Software Maintenance • Riskless Evolution: Software Environments	IS-Chap 9:	
16	**Configuration Management** • Change Management • Release Management • Version Management • Deployment Automation	IS-SE Chap 25: IS-ESP Chap 10	Assignment # 3